Classic vs. Affine DDPNM: Error and Cost across 3D Geometries

August 6, 2026

1 Summary

We compare the classic one-constant-normal-traction-per-interface **Classic-DDPNM** against the nine-mode $(\{1, s, t\} \times \{n, t_1, t_2\})$ **Affine-DDPNM** on three three-dimensional geometries, all sharing the identical tetrahedral mesh with their monolithic Taylor–Hood P2–P1 reference solution.

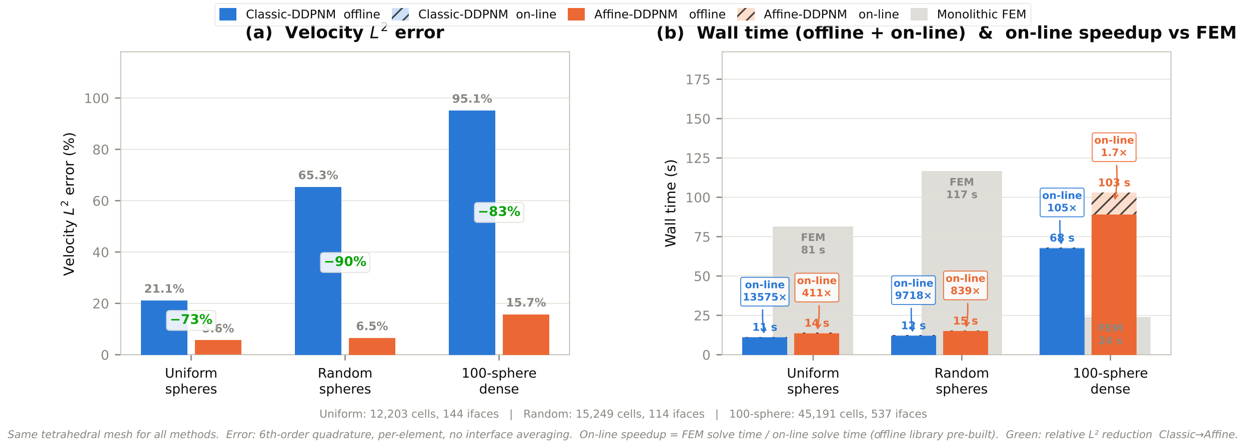

Figure 1. Comparison across three 3D geometries. (a) Velocity L^2 error relative to monolithic FEM; green labels show the relative reduction Classic → Affine. (b) Wall time breakdown: solid fill = offline local-response library; hatched overlay = on-line Schur assembly + solve + reconstruction. Grey bars = monolithic FEM reference time. Annotated on-line speedup = $t_{FEM} \,/\, t_{on-line}$, i.e. the per-boundary-condition advantage once the offline library is built. The 100-sphere benchmark uses a standard Voronoi partition; polydisperse radii ($r_{\max} - r_{\min} = 0.041 > gap/2$) cause Voronoi faces to cut through spheres, inflating both methods' errors.

2 Results

Table 1. Same-mesh error: Classic vs. Affine DDPNM.

	Classic-DDPNM				Affine-DDPNM			
Geometry	$L^2(u)$	$H^1(u)$	$L^2(p)$	Flux	$L^2(u)$	$H^1(u)$	$L^2(p)$	Flux
Uniform (27)	21.1%	44.8%	3.2%	16.5%	5.6%	21.6%	2.2%	0.3%
Random (27)	65.3%	85.9%	10.3%	72.7%	6.5%	17.6%	2.0%	2.7%
100-sphere	95.1%	115.6%	11.2%	132.2%	15.7%	30.2%	2.5%	8.7%

Table 2. Computational cost: offline (local response library), on-line (Schur assembly + solve + reconstruction), and on-line speedup vs. monolithic FEM. First-solve = offline + on-line.

	Classic-DDPNM			**Affine-DDPNM**			
Geometry	Offline	On-line	On-line ×FEM	Offline	On-line	On-line ×FEM	FEM
Uniform	11.2 s	6 ms	**13 600**×	13.7 s	0.20 s	**410**×	81.4 s
Random	12.2 s	12 ms	**9 700**×	15.1 s	0.14 s	**840**×	116.6 s
100-sphere	67.7 s	0.23 s	**105**×	89.2 s	13.8 s	**1.7**×	23.8 s

3 Key findings

1. **Affine-DDPNM reduces velocity L^2 error by 73–90%** across all geometries compared to Classic-DDPNM, with only a 23–51% increase in first-solve cost. The nine-mode interface space $\{1, s, t\} \times \{n, t_1, t_2\}$ captures in-plane linear variations that the single constant-normal mode cannot.

2. **Classic 3D DDPNM degrades severely on non-uniform geometries.** Classic velocity L^2 error grows from 21% on regular spheres to 65% on random spheres and 95% on the dense 100-sphere packing. Affine-DDPNM remains at 5.6–15.7%, with the 15.7% on the 100-sphere case dominated by the underlying Voronoi partition error (faces cutting through spheres where $r_{\max} - r_{\min} > gap/2$), not by the interface basis.

3. **On-line speedup is massive and persistent.** Classic on-line solves ($\leq$ 0.23 s) are 105–13 600× faster than monolithic FEM; Affine on-line solves ($\leq$ 14 s) are 1.7–840× faster. The on-line cost is the per-boundary-condition cost once the offline library is built—this is where DDPNM's multi-query advantage lives.

4. **First-solve speedup depends on FEM/offline ratio.** Where FEM is expensive (81–117 s on uniform/random meshes), first-solve achieves 5.9–9.6× speedup. Where FEM is cheap (23.8 s on the 45K-cell 100-sphere mesh) and offline is heavy (68–103 s, 100 pores × SuperLU per pore), first-solve is slower than FEM. This is not a failure of DDPNM—it is a mesh-size crossover effect: offline cost is a fixed entry fee that amortises as the mesh (and FEM cost) grows.

5. **The 100-sphere benchmark identifies the next bottleneck: partition geometry.** Polydisperse packings require weighted (power) Voronoi partitions to keep interfaces at true saddle points. Standard Voronoi faces cutting through spheres inflate both methods' errors; the residual 15.7% Affine error is primarily a partition-model error, not a basis error.

Mesh specifications

- **Uniform**: unit cube − $3 \times 3 \times 3$ equal spheres ($r = 0.105$), 64 maximal-ball subdomains, 144 saddle interfaces, 12,203 tetrahedra, 63,955 FEM DOFs.
- **Random**: unit cube − 27 random spheres (seed 20260804), 27 Voronoi subdomains, 114 planar Voronoi interfaces, 15,249 tetrahedra, 75,954 FEM DOFs.
- **100-sphere**: unit cube − 100 polydisperse spheres ($r \in [0.055, 0.097]$, porosity 85.6%, seed 20260806), 100 Voronoi subdomains, 537 interfaces, 45,191 tetrahedra.